

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and implemented real-time anti-abuse systems in **Node.js** and **TypeScript**, processing ~5,000 events per second with millisecond-level latency for platform security.
- Developed scalable detection pipelines using **Python** and **SQL** to transform email sending and account activity signals into automated enforcement actions, reducing manual review load by **40%**.
- Built and maintained detection rules and enforcement policies, ensuring consistent evaluation and safe, reversible updates for security controls.
- Engineered automated workflows for investigations and appeals, improving operational efficiency and handling a **35%** increase in platform usage.
- Implemented machine learning classifiers using **scikit-learn** for spam and phishing detection, validating precision and enabling automated enforcement with a **0.92 precision score** on held-out data.
- Collaborated with engineering colleagues to integrate new security signals and refine enforcement logic, contributing to a **20% reduction** in false positive incident reports.
- Containerized backend services with **Docker** and deployed to **AWS EKS**, managing configurations and ensuring reliable rollout of new security features.

SKILLS

Backend Development – Node.js, TypeScript, Go, Java, Spring Boot, REST APIs, NestJS, Hono

Databases & Messaging – PostgreSQL, Redis, AWS SQS, Kafka

Cloud & Infrastructure – AWS, AWS CDK, EC2, S3, Lambda, CloudWatch, Docker, Kubernetes, EKS

Data & ML – Python, pandas, scikit-learn, SQL, Data Warehousing, Machine Learning, ML Ops

Observability & Reliability – Datadog, Splunk, Dynatrace, Metrics, Logging, Tracing, Error Handling, Audit Trails

Security & Trust – Anti-Abuse Systems, Spam Detection, Phishing Detection, Account Takeover Prevention, Fraud Prevention, Rate Limiting, Policy Engines

PROJECTS

Real-Time Abuse Detection System (HCL Technologies)

- Developed a **Node.js** backend service to process incoming transaction events, applying real-time rules to detect and flag potentially fraudulent activities.
- Integrated with **PostgreSQL** to store user behavior patterns and historical data, enabling more accurate detection models and reducing false positives by **15%**.
- Implemented comprehensive logging and metrics using **AWS CloudWatch**, providing detailed audit trails for all detection and enforcement actions.

Automated Compliance Workflow (MSc Data Analytics Project)

- Designed and built a **Python** data pipeline to ingest and process regulatory data, transforming raw inputs into structured reports for compliance officers.
- Trained a **scikit-learn** classifier to identify non-compliant transactions, achieving an F1-score of **0.88** on a dataset of 150,000 records.
- Developed interactive **Power BI** dashboards to visualize compliance status and highlight areas requiring immediate attention.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)